Network Uncertainty and Learning from Local Observations*

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July 2026

Abstract

We study strategic interaction in networks when agents observe only their local neighborhoods and are uncertain about the broader structure in which they are embedded. Players observe their neighbors' actions through repeated play and form and revise beliefs about the network following Bayes' rule, with the network as the latent object of inference. We show that learning terminates in finite time; thereafter, play coincides with the complete-information Nash equilibrium set of the realized network, even if the network is not identified. Network identification hinges on heterogeneity in equilibrium behavior across candidate networks. If a player's own action or that of an observed neighbor distinguishes structures, the realized network is learned exactly. In the standard linear–quadratic game, equilibrium actions correspond to Katz–Bonacich centralities, so networks with distinct walk profiles are generically identified. When actions reveal players' beliefs, the speed of learning is governed by the directed diameter of the network, hence is at most linear in the number of players.

JEL Classifications: C73, D83, D85
Keywords: Bayesian learning, network games, local observation, network uncertainty

*We would like to thank Patrick Allmis, Christophe Bravard, Niloy Bose, Anindya Chakrabarti, Pritha Dev, Bhaskar Dutta, Srijita Ghosh, Ben Golub, Ahmed S. Mahmud, Matt Jackson, Luca Merlino, Debraj Ray, Ravi Ashok Satpute as well as seminar and conference participants of the University of Virginia Biocomplexity Institute, CRETE 2025, India Management Research Conference IIMA, 19th Annual Conference on Economic Growth and Development ISI New Delhi, Fifth International Conference on Issues in Economic Theory & Policy Presidency University Kolkata, Tenth Annual Conference on Network Science and Economics, the Midwest International Trade & Theory conference 2025 Spring, Southern Economic Association 95th meeting for their insightful comments and suggestions. Sudipta Sarangi would like to acknowledge the support of *Collegium de Lyon* and the *French Institutes of Advanced Study* for this research, specifically for funding received from the Horizon Europe Research and Innovation Programme (Marie Skłodowska-Curie Actions (MSCA COFUND GA-945408)).

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1 Introduction

Many economic interactions take place in networked settings. Often members of the network may not know the entire network: they only observe what happens locally and may know little about who their friends are connected to and how information reaches others. In social and economic networks for example, individuals see the behavior of their neighbors but not who influences whom beyond their immediate peers. In R&D collaboration firms may know their own collaborators, but not all the other partners of their collaborators. In organizations, workers observe close collaborators or team members without knowing the broader structure through which incentives and information propagate. Agents in these settings must form expectations about others from what they observe together with an incomplete understanding of what others observe. This raises a natural question: how do agents learn and behave when the structure of interaction itself is uncertain and must be inferred from local observations?

Existing work on Bayesian learning in repeated games sets this question aside by maintaining common knowledge of the interaction structure. Under perfect observability, all players observe every action, and learning concerns only others' behavior within a known environment (Jordan, 1991, 1995; Kalai and Lehrer, 1993a,b). Similarly, under imperfect monitoring, players observe signals of actions, but the signal structure, whether a general distribution or a fixed monitoring graph, is commonly known (Renault and Tomala, 1998, 2004). More generally, in this literature the interaction structure whether global or local is a fixed primitive and common knowledge. As a result, this literature does not address whether agents can learn the interaction structure itself from play when observation (and therefore flow of information) is decentralized.

We model this uncertainty directly. The interaction network is a primitive that jointly determines payoffs and observation, and is unknown to agents. Players hold a common prior over a finite set of candidate networks and update their beliefs using Bayes' rule based on locally observed actions generated in equilibrium play. Thus the object of learning is the interaction structure itself, rather than others' strategies or an exogenous state. This approach yields a framework in which decentralized observation and strategic behavior jointly generate the signals through which the underlying structure is inferred. It also shows how learning can result in network identification: whether agents can recover the network from play depends on how equilibrium behavior varies across candidate structures.

We study situations with a fixed but unknown network over which agents repeatedly play a game. Agents have a common prior over the set of possible networks. Each agent observes only her neighborhood and receives signals which are the actions of those with whom she interacts directly. Agents maintain Bayesian beliefs over the set of possible networks and update these beliefs based on the sequence of locally observed actions. This implies that agents do not know who observes whom or how information propagates, and must infer these from the actions of neighbors over time. At the same time, the model preserves Bayesian inference, allowing us to study learning and behavior in a setting where the information structure itself is uncertain.

Our first result characterizes the convergence of equilibrium behavior. We consider myopic agents who update beliefs about the interaction structure over time based on locally observed actions. In each period, agents choose actions that are optimal given their current beliefs, so that behavior itself reflects their inferred environment. As observations accumulate, the set of networks consistent with an agent's history shrinks monotonically and eventually stabilizes. We show that once beliefs stabilize, equilibrium play is contained within the complete-information Nash equilibrium set of the stage game under the realized interaction structure.

A key feature of this result is that convergence does not require full identification of the network. Learning operates through the interaction structure rather than directly through others' behavior, and once beliefs stabilize, any residual uncertainty about the network is payoff-irrelevant, since payoffs depend only on neighbors' actions. This distinguishes our setting from standard Bayesian learning environments. In those models, players form beliefs over others' strategies under global observability, and convergence to Nash play arises asymptotically as predictive distributions converge to those induced by realized play. In contrast, here a common prior over networks aligns beliefs about behavior conditional on the structure, so that inference is directed at the underlying structural primitive under local observation. Thus, even without full learning of the interaction structure, local observations suffice to guarantee convergence of actions.

While behavioral convergence does not require full identification of the interaction structure, learning is also feasible. It hinges on equilibrium behavior being sufficiently informative to discriminate among observationally equivalent structures. Our second result characterizes when this occurs. If the complete-information equilibrium correspondence separates candidate structures through the action of a player or one of her observed neighbors, then the realized network is identified precisely once learning stabilizes. In this case, equilibrium play eliminates all remaining observationally indistinguishable structures, so agents not only act as if the network were known, but also they learn the network structure. Network identification is thus driven by cross-structure heterogeneity in equilibrium behavior: when distinct structures induce distinct actions, observed play separates them and learning is exact.

More generally, this result highlights that network identification is driven by how equilibrium behavior maps interaction structures into actions. If two structures induce different equilibrium actions for a player or for any of her observed neighbors, then the player identifies the network once learning stabilizes. Therefore, while local observations may suffice for correct long-run behavior, full learning requires that these observations encode the structure in a way that separates all observationally equivalent networks.

We concretely illustrate this mechanism in the linear-quadratic network game of Ballester et al. (2006), where complete-information equilibrium actions admit a closed-form expression in terms of Katz-Bonacich centrality. These actions aggregate walks of all lengths originating from each player, making the mapping from networks to actions explicit. We show that if two networks differ in the number of walks of any length from a given player, then their equilibrium actions differ for almost all parameter values. Thus, whenever a player or one of her observed neighbors faces observationally indistinguishable networks with distinct walk profiles, equilibrium actions separate them once learning stabilizes. This allows agents to identify the realized network.

Lastly, we study the speed of learning. Since agents refine beliefs by eliminating structures inconsistent with observed behavior, learning concludes in finite time. We first establish a coarse bound that depends only on the size of the candidate class, then sharpen it by exploiting the local nature of observation. Because information about distant parts of the network must propagate through a sequence of interactions, the speed of learning is governed by the distance over which information must travel. This highlights a key feature of the environment. Learning is not only contingent on the accumulation of observations, but also depends on how information flows through the interaction structure.

Related literature

A natural benchmark for our paper is the Bayesian learning literature, which primarily focuses on the issue of convergence to Nash equilibrium under repeated play and perfect observability. Jordan (1991) shows that myopic Bayesian play in finite normal-form games converges almost surely to the Nash set, while Jordan (1995) extends this to forward-looking players, showing that play asymptotically converges to the Nash equilibrium of the repeated game. Kalai and Lehrer (1993a) establish the same conclusion when each player's prior assigns positive probability to strategies other players employ during the game, while Kalai and Lehrer (1993b) relax this informational restriction by showing that beliefs need only correctly predict other players' actions on the equilibrium path. We depart from this literature in two ways. First, the object being learned in our game is the interaction network rather than players' strategies. Second, in our model, convergence occurs in finite time.

A related strand examines repeated games with imperfect monitoring, where players observe noisy signals rather than others' actions. Folk theorems characterize equilibrium payoffs under public monitoring (Fudenberg et al., 1994), private monitoring with communication (Compte, 1998; Kandori and Matsushima, 1998), and without communication (Hörner and Olszewski, 2006; Sugaya, 2022). Our setting instead assumes perfect signals but an unknown structure generating them. Accordingly, we study how players learn this structure from local observations and behavior, rather than the set of sustainable equilibrium payoffs.

When the observation structure itself is uncertain, the literature typically weakens the equilibrium concept. Fudenberg and Levine (1993) introduce self-confirming equilibrium, in which beliefs about other players' strategies need only be consistent with observations along the equilibrium path. Battigalli et al. (2023) extend this notion to network games in which players observe only their realized payoffs, and characterize long-run play. Lipnowski and Sadler (2019) introduce peer-confirming equilibrium, in which a network encodes which players have correct conjectures about each other's strategies. We share the view that networks naturally restrict what each player can know about others, but retain a common prior and Bayesian updating throughout.

The network in our setting is not an exogenous informational restriction but the latent state that players learn from local observations over time.

A separate literature studies repeated games with local (rather than global) monitoring, restricting observation or communication links, but assumes this structure is common knowledge (Renault and Tomala, 1998; Ben-Porath and Kahneman, 1996; Laclau, 2014). Closer to our setting, Nava and Piccione (2014) analyze fixed networks where each player observes only her neighborhood, while Wolitzky (2013) considers networks drawn each period from a known distribution; both study which beliefs and structural features sustain cooperation. We likewise examine how beliefs over the observation structure shape long-run play, but focus on whether the structure itself can be learned. We analyze how repeated local observations refine beliefs about the structure over time and affect equilibrium behavior.

Our payoff environment belongs to the class of network games with local spillovers, in which a player's payoff depends only on her own action and the actions of their neighbors (Goyal and Joshi, 2006, 2003; Ballester et al., 2006). A separate body of work introduces incomplete information into such games in static settings. Galeotti et al. (2010) study network games in which agents only know their degree, while de Martí and Zenou (2015) analyze incomplete information about payoff-relevant parameters and characterize Bayesian-Nash equilibria. In a formation setting, Foerster et al. (2021) let some links go unobserved and characterize stable networks when players hold conjectures about the hidden structure. Beliefs there are not refined through play, whereas here repeated observation of actions resolves the uncertainty. Closest to our informational environment is Chaudhuri et al. (2026), who study static network games under network uncertainty when agents know only their local connections. We move beyond this static benchmark by making equilibrium actions observable through repeated play. The network then becomes something agents learn from the signals those actions provide, rather than a fixed object they form beliefs about in a one-shot interaction.

Our paper is also related to the literature on Bayesian learning in networks. Renault and Tomala (2004) and Li and Tan (2020) study how agents learn an unknown state under graph-based monitoring. Gale and Kariv (2003), Mueller-Frank (2013), and Huang et al. (2024) analyze how agents infer information from neighbors' actions, while Dasaratha and He (2026) quantify how network structure shapes the efficiency of signal aggregation. We differ on the object of inference: the latent state in our setting is the network itself, and informative signals take the form of endogenous equilibrium actions that reflect agents' positions in interaction structure. Convergence therefore need not take the form of consensus, as in the imitation principle of Mueller-Frank (2013); instead, the limiting outcome is contained in the complete-information Nash equilibrium set of the realized network game.

Finally, our paper is also complementary to the large literature on boundedly rational or non-Bayesian learning in networks (Bala and Goyal, 1998; Golub and Jackson, 2010; Acemoglu et al., 2010; Dasaratha and He, 2020). That literature typically asks whether decentralized updating aggregates dispersed information. Instead we maintain full Bayesian rationality and study a different friction: players observe only a local subset of equilibrium actions and use those observations to learn an initially unknown interaction structure. Our contribution is to characterize when this local inference process stabilizes, when it identifies the true network, and when the resulting behavior coincides with the complete-information Nash equilibrium of the realized network game.

The rest of the paper is structured as follows. Section 2 sets up the game. Section 3 presents the convergence and network identification results. Section 4 illustrates the results in a linear quadratic game. Section 5 characterizes the speed of learning. Section 6 concludes.

2 A Repeated Game under Network Uncertainty

We study a repeated game of strategic interaction in a finite population. In each period, a player's stage payoff depends on the actions of a subset of other players with whom she interacts, and at the end of the period she observes only the actions of those same players. The interaction structure (or network) is drawn once at the beginning of the game and remains fixed thereafter. It is not common knowledge: each player observes only her own local neighborhood and holds beliefs about the rest of the structure. Over time, players update these beliefs using signals generated by repeated play.

A distinctive feature of the model is that the unknown state is the interaction structure itself, and the signals from which players learn are the actions they observe locally. In this sense, the model combines strategic interaction with decentralized observation, where players observe only a restricted subset of actions rather than the full action profile. This stands in contrast to environments involving public-observability, in which all players observe either a common public signal or the entire profile of actions after each period. Here, learning is necessarily decentralized, since each player's information is determined by her position in the interaction structure and by the observations generated along the path of play.

Primitives

There is a finite set of players $N = \{1, \ldots, n\}$. For each $i \in N$, let $A_i \subseteq \mathbb{R}_+$ be a nonempty action set. For any subset $S \subseteq N$, let $A_S := \prod_{i \in S} A_i$, and write $A := A_N$.

Time is discrete and indexed by $\tau = 0, 1, 2, \ldots$. In each period $\tau \geq 1$, players choose actions simultaneously. Let $a_i^\tau \in A_i$ denote player i's action in period τ, and let $a^\tau = (a_i^\tau)_{i \in N} \in A$ denote the resulting action profile.

Interaction Structure

An *interaction structure* is a profile $\mathbf{g} = \big(N_i(\mathbf{g})\big)_{i \in N}$, where for each $i \in N$, $N_i(\mathbf{g}) \subseteq N \setminus \{i\}$. The set $N_i(\mathbf{g})$ represents the players with whom i interacts directly. The interaction structure determines both whose actions affect player i's payoff and whose actions player i observes at the end of each period.[1] This formulation captures environments in which the same network of local relationships determines both strategic spillovers and the information available through repeated observation.

Let $\mathcal{G}_n$ be a finite set of feasible interaction structures. At $\tau = 0$, nature draws $\mathbf{g} \in \mathcal{G}_n$ according to a commonly known prior $p \in \Delta(\mathcal{G}_n)$ with full support, and the realized structure remains fixed thereafter.

After $\mathbf{g}$ is realized, player i observes only her local interaction set $N_i(\mathbf{g})$. Her initial belief at $\tau = 1$ is therefore the prior p conditioned on the set of structures consistent with $N_i(\mathbf{g})$, that is, $p(\cdot \mid N_i(\mathbf{g}))$.

This informational environment is closely related to Chaudhuri et al. (2026), who study static network games under network uncertainty in which agents observe their

[1] More generally, one could distinguish between an influence set $N_i^u(\mathbf{g})$, which determines payoff relevance, and an observation set $N_i^z(\mathbf{g})$, which determines observed signals. We focus on the benchmark case in which these sets coincide.

local connections and update their beliefs about the rest of the network using Bayes' rule. We adopt the same informational premise but ask a different question. There, beliefs rest on local information alone; here, repeated play allows equilibrium actions to be locally observable, and these actions become the signals through which the network is progressively learned. Our focus is therefore not on equilibrium under fixed network uncertainty, but on whether and how repeated local observation resolves it over time.

Stage Payoffs

For each player i and network structure $\mathbf{g} \in \mathcal{G}_n$, stage payoffs are described by a *local spillover function*

$$v_i^{\mathbf{g}} : A_i \times A_{N_i(\mathbf{g})} \to \mathbb{R},$$

with $v_i^{\mathbf{g}} = v_i^{\mathbf{g}'}$ whenever $N_i(\mathbf{g}) = N_i(\mathbf{g}')$. Given an action profile $a \in A$, player i's payoff is

$$u_i(a, \mathbf{g}) = v_i^{\mathbf{g}}\big(a_i, a_{N_i(\mathbf{g})}\big).$$

Thus, player i's payoff depends only on her own action and the actions of her neighbors. In particular,

$$u_i(a, \mathbf{g}) = u_i(a, \mathbf{g}') \quad \text{whenever } N_i(\mathbf{g}) = N_i(\mathbf{g}').$$

This restriction implies that player i's payoff is fully determined by her neighborhood $N_i(\mathbf{g})$. The formulation nests standard linear-quadratic network games, where an agent's payoff depends on her own action and the actions of her neighbors (Ballester et al., 2006), and more generally captures local-spillover environments in which payoff-relevant interactions are mediated through direct links (Bramoullé et al., 2014). In R&D collaboration networks, for example, a firm's research payoff may depend on the R&D effort, cost reductions, or innovation outcomes of its collaborators, as in models of collaboration networks in oligopoly and R&D network formation (Goyal and Joshi, 2003, 2006; Billand et al., 2019, 2026). Similar local payoff externalities arise in network games with peer effects, workplace interactions, and local public-good environments, where an agent's payoff-relevant environment is determined by the behavior of those with whom she directly interacts (Jackson and Zenou, 2015; Calvó-Armengol et al., 2009; Lindquist et al., 2024).

We impose the following standard regularity conditions on stage payoffs.

Assumption 1. *For every player $i \in N$ and every network structure $\mathbf{g} \in \mathcal{G}_n$: (i) A_i is compact and convex, (ii) $v_i^{\mathbf{g}}$ is continuous, and (iii) for every $a_{N_i(\mathbf{g})} \in A_{N_i(\mathbf{g})}$, the map $a_i \mapsto v_i^{\mathbf{g}}(a_i, a_{N_i(\mathbf{g})})$ is concave on A_i.*

Local Observation and Private Histories

At the end of each period, player i does not observe the full action profile. Instead, she observes only her own action and the actions of the players in her neighborhood $N_i(\mathbf{g})$.[2]

Formally, player i's observation is given by a signal function

$$z_i : A \times \mathcal{G}_n \to Z_i, \quad z_i(a, \mathbf{g}) = \big(a_i, (a_j)_{j \in N_i(\mathbf{g})}\big),$$

where $Z_i := \bigcup_{S \subseteq N \setminus \{i\}} (A_i \times A_S)$. If a^τ is the action profile in period τ, the realized signal is $z_i^\tau = z_i(a^\tau, \mathbf{g})$.

Player i's private history at the beginning of period $\tau \geq 1$ is

$$h_i^\tau = \begin{cases} N_i(\mathbf{g}) & \text{if } \tau = 1, \\ \big(N_i(\mathbf{g}), z_i^1, \ldots, z_i^{\tau-1}\big) & \text{if } \tau \geq 2. \end{cases}$$

Thus, a private history consists of the observed neighborhood together with the sequence of locally observed signals. In particular, at $\tau = 1$, the history reduces to the observed neighborhood.

The set of all possible private histories for player i at period τ is

$$\mathcal{H}_i^\tau = \bigcup_{S \subseteq N \setminus \{i\}} \{S\} \times (A_i \times A_S)^{\tau-1}.$$

Under a realized network structure $\mathbf{g}$ and a profile of past play, player i's realized history satisfies $h_i^\tau \in \mathcal{H}_i^\tau$.

[2]This relates to Bayesian learning in social networks, where agents observe their neighbors' actions to learn an underlying state (Gale and Kariv, 2003; Mueller-Frank, 2013; Huang et al., 2024). There the network channels information about an exogenous state. Here it is the unknown state itself: observed actions are informative because the structure that generates them is the one that determines payoffs.

Strategies and Induced Play

A pure strategy for player i is a sequence $\sigma_i = (\sigma_i^\tau)_{\tau \geq 1}$, where

$$\sigma_i^\tau : \mathcal{H}_i^\tau \to A_i$$

assigns an action to each period-τ private history. A pure-strategy profile is denoted by $\sigma = (\sigma_i)_{i \in N}$.

Given a network structure $\mathbf{g} \in \mathcal{G}_n$ and a strategy profile σ, play is defined recursively. For each player i, let

$$h_i^1(\mathbf{g}, \sigma) := N_i(\mathbf{g}).$$

For each period $\tau \geq 1$, define

$$a_i^\tau(\mathbf{g}, \sigma) = \sigma_i^\tau\big(h_i^\tau(\mathbf{g}, \sigma)\big), \qquad a^\tau(\mathbf{g}, \sigma) = \big(a_k^\tau(\mathbf{g}, \sigma)\big)_{k \in N} \qquad ,$$

and

$$z_i^\tau(\mathbf{g}, \sigma) = z_i\big(a^\tau(\mathbf{g}, \sigma), \mathbf{g}\big).$$

The private history then evolves according to

$$h_i^{\tau+1}(\mathbf{g}, \sigma) = \big(N_i(\mathbf{g}), z_i^1(\mathbf{g}, \sigma), \ldots, z_i^\tau(\mathbf{g}, \sigma)\big).$$

Thus, for each $\mathbf{g}$ and σ, the model induces a unique sequence

$$\big\{(h^\tau(\mathbf{g}, \sigma), a^\tau(\mathbf{g}, \sigma), z^\tau(\mathbf{g}, \sigma))\big\}_{\tau \geq 1}.$$

Distinct interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$ may generate identical private histories for player i under σ, and are therefore indistinguishable from her perspective. This observation motivates the following definition.

Definition 1. *Fix a strategy profile σ, a player i, and a period $\tau \geq 1$. Two interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$ are observationally equivalent for player i at period τ under σ, written $\mathbf{g} \sim_{i,\tau} \mathbf{g}'$, iff $h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau(\mathbf{g}', \sigma)$. We denote by $[\mathbf{g}]_{i,\tau}$ the corresponding equivalence class.*[3]

[3] The dependence of the relation $\sim_{i,\tau}$ and $[\mathbf{g}]_{i,\tau}$ on σ is suppressed whenever the strategy profile is fixed.

Beliefs

Fix a player i and a period $\tau \geq 1$. Since player i's private history at the beginning of period τ is determined by the realized interaction structure together with play in periods $1, \ldots, \tau - 1$, the prior p and the strategy profile σ induce a probability distribution over $\mathcal{H}_i^\tau$. For any history $h_i^\tau \in \mathcal{H}_i^\tau$, define

$$P_{i,\sigma}(h_i^\tau) := \sum_{\mathbf{g} \in \mathcal{G}_n} p(\mathbf{g}) \, \mathbb{I}\{h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau\}.$$

Thus, $P_{i,\sigma}(h_i^\tau)$ is the ex ante probability that player i's realized private history at the beginning of period τ is h_i^τ.

The strategy profile σ therefore induces, for each player i and each period τ, a posterior belief over interaction structures at every history in $\mathcal{H}_i^\tau$. Let the induced belief system be denoted by

$$\mu_\sigma = \left(\mu_{i,\sigma}^\tau\right)_{i \in N, \ \tau \geq 1},$$

where

$$\mu_{i,\sigma}^\tau(\,\cdot \mid h_i^\tau) \in \Delta(\mathcal{G}_n)$$

denotes player i's posterior at history $h_i^\tau \in \mathcal{H}_i^\tau$. At every history such that $P_{i,\sigma}(h_i^\tau) > 0$, these beliefs are obtained by Bayes' rule:

$$\mu_{i,\sigma}^\tau(\mathbf{g} \mid h_i^\tau) = \frac{p(\mathbf{g}) \, \mathbb{I}\{h_i^\tau(\mathbf{g}, \sigma) = h_i^\tau\}}{P_{i,\sigma}(h_i^\tau)}, \qquad \text{for all } \mathbf{g} \in \mathcal{G}_n.$$

At histories $h_i^\tau \in \mathcal{H}_i^\tau$ such that $P_{i,\sigma}(h_i^\tau) = 0$, beliefs may be specified arbitrarily.

Define the path of play of player i under σ by

$$\mathcal{P}_i(\sigma) := \bigcup_{\tau \geq 1} \left\{h_i^\tau \in \mathcal{H}_i^\tau : P_{i,\sigma}(h_i^\tau) > 0\right\},$$

and define

$$\mathcal{P}(\sigma) := \bigcup_{i \in N} \mathcal{P}_i(\sigma)$$

as the path of play induced by σ.

Because the prior p has full support, for every $\mathbf{g} \in \mathcal{G}_n$, every player i, and every period

τ, the induced history $h_i^\tau(\mathbf{g}, \sigma)$ satisfies

$$P_{i,\sigma}\big(h_i^\tau(\mathbf{g}, \sigma)\big) \geq p(\mathbf{g}) > 0.$$

Hence, every history generated by some interaction structure under σ lies on the path of play.

Equilibrium

Players are myopic in the sense that, in each period, they choose actions to maximize current expected payoff given their posterior beliefs. This corresponds to the dynamic counterpart of Jordan (1991), where the unknown state is the network structure $\mathbf{g}$ and beliefs are updated through locally observed signals z_i^τ. In contrast to models with forward-looking behavior (e.g., Kalai and Lehrer (1993a); Jordan (1995)), repeated interaction here serves to generate information rather than intertemporal incentives.

Definition 2. *A strategy profile σ^* is a myopic Bayesian equilibrium if, for every player i, every period $\tau \geq 1$, and every history $h_i^\tau \in \mathcal{P}_i(\sigma^*)$,*

$$\sigma_i^{*\tau}(h_i^\tau) \in \arg\max_{a_i \in A_i} \sum_{\mathbf{g} \in \mathcal{G}_n} \mu_{i,\sigma^*}^\tau(\mathbf{g} \mid h_i^\tau)\, u_i\left(a_i,\, \sigma_{-i}^{*\tau}\big(h_{-i}^\tau(\mathbf{g}, \sigma^*)\big);\, \mathbf{g}\right),$$

where $h_{-i}^\tau(\mathbf{g}, \sigma^*) = \big(h_j^\tau(\mathbf{g}, \sigma^*)\big)_{j \neq i}$.

The equilibrium condition constrains behavior only on the path of play $\mathcal{P}(\sigma^*)$. At histories $h_i^\tau \notin \mathcal{P}_i(\sigma^*)$, the strategy $\sigma_i^{*\tau}(h_i^\tau)$ is unconstrained, and two strategy profiles that agree on $\mathcal{P}(\sigma^*)$ are equivalent. Under Assumption 1, a myopic Bayesian equilibrium exists[4].

The myopic equilibrium abstracts from intertemporal incentives and, in particular, from the use of actions to influence future beliefs or continuation play. When players maximize current expected payoffs, they form beliefs given current action. Given these beliefs, their actions depend only on what they have inferred about the interaction structure, and so convey that information to those who observe them. This is essential in our setting, where players learn the structure precisely by observing their neighbors' actions. Under forward-looking behavior, actions could instead be chosen to influence

[4] In each period, beliefs are pinned down by Bayes' rule on the path of play, and the per-period objective is continuous in a_i and concave on the compact, convex set A_i, so a maximizer exists.

others' beliefs or subsequent play, and would no longer serve as reliable signals of what each player has learned. This feature is central to our analysis, as it isolates the informational content of play and underlies our identification and convergence results. In this sense, the equilibrium concept plays a role analogous to the best-response dynamics studied in the Bayesian learning literature (Jordan, 1991, 1995; Kalai and Lehrer, 1993b,a), adapted to an environment in which the observation structure is endogenous and unknown. The concept is also related to self-confirming equilibrium (Battigalli et al., 2023; Fudenberg and Levine, 1993), with the distinction that beliefs here are fully disciplined by a common prior and Bayes' rule at all on-path histories.[5]

3 Convergence and Identification

This section studies behavior and information once learning stabilizes. Two distinct findings arise. First, agents may eventually behave as if they know the true interaction structure, in the sense that equilibrium play coincides with the complete-information Nash equilibrium under the realized structure. This need not imply that they identify the structure itself. Second, agents may or may not be able to identify the realized interaction structure. We now show that the first property holds quite generally, whereas the second depends on whether equilibrium actions are sufficiently informative to distinguish between observationally equivalent structures.

Fix a myopic Bayesian equilibrium σ^*. Throughout this section, all histories, observational-equivalence classes, and induced actions are defined relative to σ^*. For each player $i \in N$, period $\tau \geq 1$, and network structure $\mathbf{g} \in \mathcal{G}_n$, let $[\mathbf{g}]_{i,\tau}$ denote the observational-equivalence class of $\mathbf{g}$ for player i at period τ, that is,

$$[\mathbf{g}]_{i,\tau} = \{\mathbf{g}' \in \mathcal{G}_n : h_i^\tau(\mathbf{g}', \sigma^*) = h_i^\tau(\mathbf{g}, \sigma^*)\}.$$

Because the prior p has full support, at any on-path history $h_i^\tau(\mathbf{g}, \sigma^*)$, the support of player i's posterior coincides with $[\mathbf{g}]_{i,\tau}$.

[5]With forward-looking players, actions could manipulate beliefs or sustain history-dependent play. This introduces signaling and punishment motives that confound their informational role. Section 3.1 shows our results extend under restrictions that remove such effects, so myopia serves as a tractable benchmark that isolates learning.

Throughout, for each $\mathbf{g} \in \mathcal{G}_n$, let $NE(\mathbf{g}) \subseteq A$ denote the set of pure strategy Nash equilibria of the complete information stage game induced by $\mathbf{g}$ and let $a^c(\mathbf{g})$ be a generic element of $NE(\mathbf{g})$ with $a_i^c(\mathbf{g})$ denoting its i^{th} component. For each player $i \in N$, write $NE_i(\mathbf{g}) := \{a_i(\mathbf{g}) : a(\mathbf{g}) \in NE(\mathbf{g})\}$ for the projection of the complete-information Nash equilibrium set under $\mathbf{g}$ onto player i's action space.

3.1 Convergence to Stage-Game Equilibrium

Observation in this environment is local: players observe only the actions of those in their interaction neighborhoods. We show that, once learning stabilizes, this local information is nonetheless sufficient to align equilibrium behavior with the complete-information benchmark.

We begin by formalizing when learning ends.

Definition 3. *Learning ends at period τ^* if, for all $\tau \geq \tau^*$,*

$$[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,\tau^*} \qquad \text{for all } i \in N \text{ and } \mathbf{g} \in \mathcal{G}_n.$$

After τ^*, no player rules out further interaction structures on the basis of observed play. This does not require that players identify the true structure, only that the equilibrium path generates no additional eliminations. In particular, if $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$ for some $\tau \geq \tau^*$, then $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$, so $\mathbf{g}$ and $\mathbf{g}'$ generate the same next-period signal for player i.

Lemma 1. *For every myopic Bayesian equilibrium σ^*, there exists a finite period τ^* such that learning ends, i.e. for all $i \in N$ and $\mathbf{g} \in \mathcal{G}_n$,*

$$[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,\tau^*} \qquad \forall \tau \geq \tau^*.$$

This proof and all other proofs are relegated to the Appendix. We now state the main result. It shows that once learning stabilizes, any residual uncertainty becomes payoff-irrelevant along the equilibrium path.

Theorem 1. *Let $\mathbf{g} \in \mathcal{G}_n$ be the realized interaction structure. If learning ends at τ^* under σ^*, then*

$$a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g}) \qquad \forall \tau \geq \tau^*.$$

The logic is straightforward. Once learning ends, all interaction structures in $[\mathbf{g}]_{i,\tau^*}$ generate the same payoff-relevant local observations at the realized history. Residual uncertainty is therefore irrelevant for player i's best-response problem.

Theorem 1 is our core behavioral implication. Even if players may fail to identify the realized interaction structure ($|[\mathbf{g}]_{i,\tau^*}|$ may exceed one) they behave as if it were common knowledge. Uncertainty about the network structure thus becomes behaviorally irrelevant in the long run. All of this rests on myopia. Were players forward-looking, an action could be chosen for its effect on others' beliefs or on future play, and the correspondence between learning and complete-information behavior in Theorem 1 need no longer hold. We take this up at the end of the section, and find that it is restored under a mild condition.

The theorem departs from classical Bayesian learning results such as Kalai and Lehrer (1993a) and Jordan (1995), in two ways. First, we replace global observability with strictly local observability. Second, we obtain finite-time rather than asymptotic convergence.

The result also contrasts with the imitation principle in Mueller-Frank (2013). In that setting, agents observe neighbors' actions to learn about a common payoff-relevant state. Their payoffs, however, do not directly depend on those neighbors' actions. Since agents face the same decision problem, an agent who observes a neighbor can copy the neighbor's action, and this disciplines long-run disagreement. Our model differs because the realized network is the payoff-relevant state. Neighbors' actions enter payoffs through local strategic spillovers, so players in different network positions generally face different best-response problems. Therefore, learning need not produce imitation or consensus. Once learning stabilizes, play enters the set of complete-information Nash equilibria of the realized network game. Unlike the consensus outcome under the imitation principle, this profile is generally heterogeneous across players, since agents in different network positions face different best-response problems. Moreover, note that when the stage game admits multiple equilibria, the realized profile need not be unique, but it remains a complete-information Nash equilibrium of the realized network in every period.

Consider, then, forward-looking play. Suppose σ^* is a perfect Bayesian equilibrium. Let τ^* denote the stabilization time from Lemma 1. Although $[g]_{i,\tau}$ remains constant for all $\tau \geq \tau^*$, a deviation may affect continuation values through its impact on beliefs or prescribed future play, even if it does not alter the observational-equivalence class along the equilibrium path. Residual uncertainty within $[g]_{i,\tau^*}$ may therefore be irrelevant for stage payoffs but relevant for continuation values, so equilibrium behavior need not coincide with the complete-information benchmark.

Remark. *With forward-looking players, Theorem 1 continues to hold whenever unilateral deviations do not affect posterior beliefs or continuation play.*

This is precisely the feedback that myopia targets. Suppose that for all $\tau \geq \tau^*$, continuation strategies depend only on $[g]_{i,\tau^*}$ and unilateral deviations do not affect posterior beliefs or continuation play, then continuation payoffs are invariant to deviations. Each player's problem reduces to a static best-response problem given beliefs over $[g]_{i,\tau^*}$, and

$$a_i^\tau(g,\sigma^*) \in \mathrm{BR}_i(a_{-i}^\tau(g,\sigma^*); g) \quad \text{for all } i,$$

so that $a^\tau(g,\sigma^*)$ coincides with the complete-information Nash equilibrium.

3.2 Perfect Learning

Theorem 1 shows that long-run behavior coincides with the complete-information benchmark. We now ask when a player fully identifies the realized network.

Definition 4. *Fix $\mathbf{g} \in \mathcal{G}_n$. Learning is* perfect *for player i if*

$$[\mathbf{g}]_{i,\tau^*} = \{\mathbf{g}\}.$$

Perfect learning is stronger than behavioral convergence. It requires that equilibrium behavior be sufficiently informative to eliminate all incorrect interaction structures, and not merely to pin down equilibrium actions. The next theorem shows when this can happen.

Theorem 2. *Fix $\mathbf{g} \in \mathcal{G}_n$ and suppose learning ends at τ^* under σ^*. Fix $i \in N$. If either of the following conditions holds, then learning is perfect for player i:*[6]

[6] This separation condition is a sufficient identification condition rather than a restriction. Under uniqueness, it reduces to injectivity of the map $\mathbf{g}' \mapsto a_k^c(\mathbf{g}')$. With multiple equilibria, it requires the projected equilibrium sets to be disjoint across structures. In environments with rich action spaces, observed actions may reveal beliefs or informational states; see Arieli and Mueller-Frank (2017) and Molavi et al. (2018). In the linear-quadratic network game (Ballester et al., 2006), equilibrium actions are generically one-to-one in the network's walk profile, as we show below.

(i) $NE_i(\mathbf{g}') \cap NE_i(\mathbf{g}'') = \emptyset$ *for every distinct* $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$.

(ii) *there exists* $j \in N_i(\mathbf{g})$ *such that* $NE_j(\mathbf{g}') \cap NE_j(\mathbf{g}'') = \emptyset$ *for every distinct* $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$.

Theorem 2 strengthens behavioral convergence to full identification. An observed action is always informative: it refines a player's beliefs by ruling out networks inconsistent with it. Identification requires more: that the refinement process continues until a single network remains. This happens when distinct structures induce distinct actions, so heterogeneity in actions across structures is what pins the network down. The two conditions require this heterogeneity in an action i can observe. Condition (i) requires it of i's own action, so i pins the structure down from what she plays. Condition (ii) requires it of a neighbor: some $j \in N_i(\mathbf{g})$ acts differently under each network, so watching j pins it down. When actions are homogeneous across structures, refinement stalls short of identification, even though behavior still converges correctly by Theorem 1.

Together, Theorems 1 and 2 define the limits of decentralized learning here. Local observations suffice for correct long-run behavior: once learning stabilizes, agents act as if the interaction structure were known, despite incomplete network knowledge. However, identifying the structure itself requires a stronger condition, that equilibrium behavior be informative enough to distinguish observationally equivalent structures. Thus behavior and knowledge may diverge since agents can act optimally relative to the true structure without ever learning it. This distinction reflects a fundamental feature of environments with endogenous and locally generated information showing that learning is constrained not only by what is observed, but also by how equilibrium behavior encodes underlying primitives. In this sense, information along the path of play is determined jointly by the interaction structure and the equilibrium mapping from structure to actions.

4 Linear Quadratic Network Games

We illustrate the general convergence results for the workhorse linear-quadratic network game to provide a transparent interpretation of both convergence and identification. In this environment, equilibrium actions admit a closed-form representation that makes explicit how local interactions aggregate information about the global structure.

4.1 Environment

For each player $i \in N$, let $A_i = \mathbb{R}$ and define stage payoffs by

$$u_i(a, \mathbf{g}) = a_i - \frac{1}{2}a_i^2 + \lambda a_i \sum_{j \in N} g_{ij}(\mathbf{g})a_j. \tag{1}$$

where $g_{ij}(\mathbf{g}) = 1$ if $j \in N_i(\mathbf{g})$, and 0 otherwise. This is the linear-quadratic network game of Ballester et al. (2006), among the most widely studied models in the networks literature. Player i chooses an action that trades off a quadratic cost against complementarities with the actions of her neighbors. The interaction structure affects payoffs only through the set of neighbors $N_i(\mathbf{g})$, so the local-spillovers property of the general model is preserved.

Define $G(\mathbf{g}) = [g_{ij}(\mathbf{g})]_{1 \leq i,j \leq n}$ as the adjacency matrix representation of the interaction structure. Let $\bar{\rho} := \max_{\mathbf{h} \in \mathcal{G}_n} \rho(G(\mathbf{h}))$, where $\rho(G(\mathbf{h}))$ is the spectral radius of the corresponding adjacency matrix, and suppose $0 < \lambda < 1/\bar{\rho}$. Then, for any $\mathbf{g} \in \mathcal{G}_n$, the unique pure Nash equilibrium of the complete-information stage game is given by

$$a^c(\mathbf{g}) = \big(I - \lambda G(\mathbf{g})\big)^{-1}\mathbf{1}, \tag{2}$$

which corresponds to the Katz–Bonacich centrality vector of the network $\mathbf{g}$.

We now connect this structure to the general results. Let $\mathbf{g}$ be the realized interaction structure, and suppose learning ends at period τ^* under σ^*. By Theorem 1, equilibrium play converges to the complete-information Nash equilibrium under $\mathbf{g}$. Combining this with (2) yields:

Corollary 1. *Suppose payoffs are given by (1). If learning ends at period τ^* under σ^* and the realized interaction structure is $\mathbf{g}$, then the equilibrium action profile coincides with the corresponding Katz–Bonacich centrality vector of the network $\mathbf{g}$, i.e.*

$$a_i^\tau(\mathbf{g}, \sigma^*) = \Big[\big(I - \lambda G(\mathbf{g})\big)^{-1}\mathbf{1}\Big]_i \qquad \forall i \in N,\ \forall \tau \geq \tau^*.$$

The corollary shows that even though players initially observe only their local neighborhoods and must infer the rest of the network indirectly, equilibrium play aggregates information across the network. Once learning stabilizes, each player behaves exactly as if the entire interaction structure were known. Any remaining uncertainty is confined to structures that are observationally and behaviorally indistinguishable.

4.2 Identification via Network Walks

The centrality representation also clarifies when equilibrium actions identify the underlying interaction structure. Expanding (2) as a Neumann series, we obtain

$$a^c(\mathbf{g}) = \sum_{k=0}^{\infty} \lambda^k G(\mathbf{g})^k \mathbf{1}. \tag{3}$$

For each player i and $k \geq 0$, define

$$W_i^k(\mathbf{g}) := \left[G(\mathbf{g})^k \mathbf{1}\right]_i.$$

Because $G(\mathbf{g})$ is the adjacency matrix, $W_i^k(\mathbf{g})$ counts the number of walks of length k that originate at player i. Thus,

$$a_i^c(\mathbf{g}) = \sum_{k=0}^{\infty} \lambda^k W_i^k(\mathbf{g}),$$

so equilibrium actions encode the entire walk structure of the network around each player.

This is where behavior meets identification. Player i's action aggregates walks of every length, so it reflects the network's higher-order structure, not just her direct links. Two networks that differ at any walk length must then differ in her action, for almost every λ[7]. Behavior separates them, and identification follows.

Proposition 1. *Fix $i \in N$ and two networks $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$. Suppose there exists $k \geq 1$ such that $W_i^k(\mathbf{g}) \neq W_i^k(\mathbf{g}')$. Then $a_i^c(\mathbf{g}) \neq a_i^c(\mathbf{g}')$ for almost all values of λ satisfying $0 < \lambda < 1/\max\{\rho(G(\mathbf{g})), \rho(G(\mathbf{g}'))\}$.*

[7] In our context "almost all λ" means for all admissible λ outside a set of Lebesgue measure zero. In each of our results this exceptional set is finite and it consists of the roots of a polynomial in λ so the property holds for all but finitely many values.

The intuition follows directly from (3). Equilibrium actions aggregate walk counts of all lengths. If two networks differ at any walk length, then their corresponding power series generically differ, and hence define distinct equilibrium actions for almost all admissible values of λ.

An additional implication of Proposition 1 is how sharply equilibrium actions distinguish networks in the linear-quadratic game. Although each network yields a unique Nash equilibrium, for almost all λ, networks with different walk profiles generate distinct equilibrium action profiles. Hence, equilibrium behavior is generically one-to-one in the network's walk profile. This reflects the model's core structure: equilibrium actions encode the network via walks, yielding a strong form of behavioral identifiability and the following corollary.

Corollary 2. *Fix $\mathbf{g} \in \mathcal{G}_n$ and $i \in N$. Suppose some $j \in \{i\} \cup N_i(\mathbf{g})$ can distinguish every pair of interaction structures in $\mathcal{G}_n$ through its walk profile, i.e., $\mathbf{g}' \neq \mathbf{g}'' \in \mathcal{G}_n$ implies $W_j^k(\mathbf{g}') \neq W_j^k(\mathbf{g}'')$ for some $k \geq 1$. Then, if learning terminates at period τ^* under σ^*, player i achieves perfect learning for almost all admissible values of λ.*

The linear-quadratic environment provides a concrete illustration of the general results. Information about the network propagates through walks, and equilibrium actions aggregate these effects via the Katz–Bonacich representation; once learning stabilizes, behavior reflects the entire upstream network relevant for payoffs. This is Theorem 1, local observation and equilibrium response suffice to recover the payoff-relevant structure. Proposition 1 adds that identification turns on whether these effects uniquely encode the network. When walk profiles differ, actions separate the structures for almost all λ, and learning is perfect. Convergence thus reflects how payoff-relevant influence propagates through the network; identification, whether that influence generates distinct walk profiles.

Note that the linear-quadratic utility is used only for the closed-form Katz–Bonacich representation and the walk-based identification result; convergence itself does not require interior equilibria. It applies to any network game with local payoff spillovers, including the linear-best-reply class of Bramoullé et al. (2014), which comprises quadratic games, local public-good games and R&D networks, with multiplicity in equilibria and corner solutions. In all of these, once learning terminates under a myopic Bayesian equilibrium, the realized profile is a complete-information Nash equilibrium of the realized network. Corner solutions and multiplicity therefore do not break the bridge from learning to complete-information behavior.

5 Speed of Learning

Section 3 established two conclusions. First, for any fixed myopic Bayesian equilibrium, each player's observational-equivalence class shrinks only through elimination and therefore stabilizes in finite time. Second, once learning stabilizes, equilibrium play coincides with the complete-information Nash equilibrium of the realized interaction structure. We now ask a natural follow up question: *how long can the learning process last?*

Fix an arbitrary myopic Bayesian equilibrium σ^*. By Lemma 1, the earliest period at which learning ends under σ^* is well defined. Denote this time by

$$\tau^*(\sigma^*) := \min\Big\{\tau \geq 1 : [\mathbf{g}]_{i,\tau+1} = [\mathbf{g}]_{i,\tau} \text{ for all } i \in N,\ \mathbf{g} \in \mathcal{G}_n\Big\}.$$

Because the classes $[\mathbf{g}]_{i,\tau}$ are non-increasing in τ, equality between periods τ and $\tau+1$ is equivalent to stabilization from period τ onward.

Two forces determine $\tau^*(\sigma^*)$. The first is purely combinatorial and uses only the finiteness of $\mathcal{G}_n$. The second reflects the topology of local observation: information propagates only along directed paths in the interaction structure, so that learning speed is governed by the depth of these paths.

5.1 A Coarse Bound from Finite Refinements

For each player i and period $\tau \geq 1$, let

$$\Pi_i^\tau(\sigma^*) := \Big\{[\mathbf{g}]_{i,\tau} : \mathbf{g} \in \mathcal{G}_n\Big\}, \qquad b_i^\tau(\sigma^*) := \big|\Pi_i^\tau(\sigma^*)\big|.$$

The collection $\Pi_i^\tau(\sigma^*)$ is the partition of $\mathcal{G}_n$ induced by player i's period-τ private histories and provides a representation of player i's information at period τ. Each element collects interaction structures that are observationally indistinguishable given i's private history. As learning proceeds, this partition becomes finer since new observations refine indistinguishable structures into smaller sets. The number of cells

$b_i^\tau(\sigma^*)$ therefore proxies the amount of information player i has acquired. Bounding the speed of learning reduces to bounding how many such refinements can occur before the partition stabilizes.

From the definition of private histories, for every $i \in N$, $\tau \geq 1$, and $\mathbf{g} \in \mathcal{G}_n$, we have that

$$[\mathbf{g}]_{i,\tau+1} \subseteq [\mathbf{g}]_{i,\tau}.$$

Thus, $\Pi_i^{\tau+1}(\sigma^*)$ is a refinement of $\Pi_i^\tau(\sigma^*)$, and the sequence

$$b_i^1(\sigma^*),\, b_i^2(\sigma^*),\, b_i^3(\sigma^*), \ldots$$

is weakly increasing.

Theorem 3. *Fix a myopic Bayesian equilibrium σ^*. Suppose that σ^* satisfies the following condition: $\Pi_i^{\tau+1}(\sigma^*) = \Pi_i^\tau(\sigma^*) \Rightarrow a_i^{\tau+1}(\mathbf{g}, \sigma^*) = a_i^\tau(\mathbf{g}, \sigma^*)$, for every $i \in N$ and every $\mathbf{g} \in \mathcal{G}_n$. Then, $\tau^*(\sigma^*) \leq 1 + n(|\mathcal{G}_n| - 1)$.*

Theorem 3 follows from a simple counting argument. Each strict refinement of $\Pi_i^\tau(\sigma^*)$ increases $b_i^\tau(\sigma^*)$ by at least one, and for each player i, the number of such increases is bounded by $|\mathcal{G}_n| - 1$. Aggregating across players yields the stated bound.

This bound is coarse and depends only on the size of the candidate class. It does not exploit the structure of local observation and treats all refinements as equally informative. Information in our environment, however, propagates through the interaction structure along constrained paths. The next subsection uses this underlying structure to obtain sharper bounds.

The stationarity condition of Theorem 3 is used only to convert a bound on the number of strict refinements into a bound on calendar time. Without this condition, the finite state space still bounds the number of times information partitions can be strictly refined, but it does not bound when those refinements occur. In particular, a nonstationary equilibrium selection could keep the same information partition for several periods and reveal already-held information only at a later calendar date.[8]

[8] For example, suppose player j can distinguish two interaction structures $\mathbf{g}$ and $\mathbf{g}'$, while player i cannot, and i observes j. If two actions are payoff-equivalent for j, then an equilibrium selection may prescribe the same action for j under $\mathbf{g}$ and $\mathbf{g}'$ for τ periods and then prescribe different actions in period $\tau + 1$. Player i's partition would not refine before period $\tau + 1$, but would refine afterward.

5.2 A Distance Bound under Action Separation

A sharper bound can be found once we exploit the fact that learning in our model is local. Under local observation, player i never observes the entire interaction structure directly. Instead, she learns about distant parts of the network only indirectly through the actions of the players she observes. This induces a propagation structure: information about a remote node must first affect that node's own history, then that node's action, then a downstream player's signal, and so on. Information can thus travel at most one edge per period.

To convert this propagation structure into a sharper bound, however, it is not enough that actions respond to histories. It is important that observed actions reveal the histories that generated them. Otherwise, information may be generated upstream, but fail to be decoded by downstream players.

To formalize the resulting propagation structure further, we orient links as follows

$$j \to i \qquad \Longleftrightarrow \qquad j \in N_i(\mathbf{g}).$$

Thus, information flows toward the observer.

For a realized network structure $\mathbf{g}$, define the directed distance from j to i by

$$\mathrm{dist}_{\mathbf{g}}(j,i) := \min\Big\{\ell \in \mathbb{N}_0 : \exists \text{ a directed path } j \to \cdots \to i \text{ of length } \ell\Big\},$$

with the convention $\mathrm{dist}_{\mathbf{g}}(j,i) = \infty$ if no such path exists. For $r \geq 0$, define the upstream r-neighborhood of i by

$$N_i^r(\mathbf{g}) := \{\, j \in N : \mathrm{dist}_{\mathbf{g}}(j,i) \leq r \,\}.$$

Hence,

$$N_i^0(\mathbf{g}) = \{i\}, \qquad N_i^1(\mathbf{g}) = \{i\} \cup N_i(\mathbf{g}).$$

For $r \geq 1$, define the rooted upstream radius-r structure of player i by

$$R_i^r(\mathbf{g}) := \big(N_j(\mathbf{g})\big)_{j \in N_i^{r-1}(\mathbf{g})}.$$

This object records the incoming neighborhoods of all nodes whose directed distance to i is at most $r-1$. Equivalently, it is the subset of the interaction structure that can possibly affect player i's observations within r rounds of play. Nodes outside this rooted upstream neighborhood are too far away to have influenced what player i has observed by time r. The next lemma summarizes the informational content of a player's history.

Lemma 2. *Fix a myopic Bayesian equilibrium σ^*. For every $i \in N$, $\tau \geq 1$, and $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$,*

$$R_i^\tau(\mathbf{g}) = R_i^\tau(\mathbf{g}') \Rightarrow h_i^\tau(\mathbf{g}, \sigma^*) = h_i^\tau(\mathbf{g}', \sigma^*). \tag{4}$$

If in addition, equilibrium actions separate informational classes, that is,

$$[\mathbf{g}]_{i,\tau} \neq [\mathbf{g}']_{i,\tau} \quad \Longrightarrow \quad a_i^\tau(\mathbf{g}, \sigma^*) \neq a_i^\tau(\mathbf{g}', \sigma^*) \qquad \textit{for all } i \textit{ and } \tau$$

then the converse of (4) also holds.

Lemma 2 has a simple economic interpretation. The separation condition requires a player's action to vary across her informational classes, so that distinct classes have a distinct impact on what she does.[9] Behavior then does not merely respond to local information but also transmits it, so that observing a neighbor's action is equivalent to learning what she has inferred so far. By time τ, player i can therefore learn only the part of the network whose consequences could have reached her through $\tau - 1$ rounds of locally observed play. When actions are separating no information is lost, so this bound is exact: player i's history at time τ is informationally equivalent to knowing the entire upstream portion of the network upto radius τ. The relevant state variable for time-τ learning is thus not the whole interaction structure, but the rooted upstream τ-neighborhood. Like the conditions for perfect learning, separation is a heterogeneity requirement, but of a different kind. There, identification depended on heterogeneity in behavior across structures; here, what matters is heterogeneity in a player's own behavior across her informational classes. A player whose action varies with what she has inferred transmits that inference to those who observe her, whereas one who acts identically across distinct classes leaves sign of what separates them.

This equivalence yields a sharper bound on the speed of learning. For every connected

[9] It requires equilibrium actions to reveal current informational states along the equilibrium path. This is related to the idea that actions can encode beliefs or private information when the action space is sufficiently rich; see, for example, Arieli and Mueller-Frank (2017).

network $\mathbf{g}$, define its directed diameter by

$$\mathrm{diam}(\mathbf{g}) := \max_{i,j \in N} \big\{ \mathrm{dist}_{\mathbf{g}}(j,i) : \mathrm{dist}_{\mathbf{g}}(j,i) < \infty \big\},$$

and define the maximal directed diameter across the candidate class by

$$d(\mathcal{G}_n) := \max_{\mathbf{g} \in \mathcal{G}_n} \mathrm{diam}(\mathbf{g}).$$

Proposition 2. *Suppose equilibrium actions separate informational classes. Then*

$$\tau^*(\sigma^*) \leq d(\mathcal{G}_n) + 1.$$

Proposition 2 shows that the speed of learning is governed by the topology of information flow. When the candidate class has small diameter, all strategically relevant uncertainty is exhausted quickly because information propagates through the entire network in only a few rounds. When the diameter is large, learning is slow because information from distant actions must pass through many intermediate neighbors and be reflected in their actions before reaching the agent. The proposition therefore shows that local observation creates a propagation friction, while equilibrium separation determines whether that local information can be decoded without loss. Because the bound is governed by diameter rather than by the number of players, learning is rapid whenever the diameter is short, as in small-world networks (Watts and Strogatz, 1998). The number of players nonetheless enters indirectly, since it bounds the diameter itself. A direct corollary is that the learning time is at most linear in the number of players.

Corollary 3. *Suppose equilibrium actions separate informational classes. Then*

$$\tau^*(\sigma^*) \leq n.$$

In summary, the two bounds are complementary. While Theorem 3 is general, resulting from the finite number of possible interaction structures, it abstracts from the structure of information flow. Proposition 2 exploits this structure to provide sharper bounds, but requires that equilibrium actions separate informational classes, ensuring observed behavior fully reveals current informational states.

6 Conclusion

This paper studies strategic interaction when the underlying network of interactions is unknown to agents. We show that agents can use local observations of behavior to form and update beliefs over candidate networks within a standard Bayesian framework. In finite time, equilibrium behavior enters the set of complete-information Nash equilibria of the realized stage game, even when the structure is not identified. Identification itself requires heterogeneity in equilibrium behavior across candidate networks, and in the linear-quadratic class this heterogeneity takes the form of distinct walk profiles. Behavioral convergence and structural learning are thus distinct properties, with the latter requiring that equilibrium actions vary sufficiently across networks.

This paper also has interesting implications for some other areas. The first relates to econometric identification of peer effects from observed behavior. Our results provide a way to understand how observed behavior can be used to derive heterogeneity conditions that can be subsequently used for identifying peer effects from reduced-form outcomes. Recall that the realized structure can be recovered from observed actions depending on whether the equilibrium mapping is separating across candidate networks or not. The walk-profile characterization in the linear-quadratic class gives a concrete form for this condition. Our findings also provide a note of caution regarding the time of data collection for network models. Typically, these models assume equilibrium behavior and often work with the linear best-response equation. This assumption of equilibrium behavior requires that the network has been in place for a sufficiently long period of time to ensure that the members are playing best responses. The second implication is for policy when players are parts of networks they do not fully observe. Local observation suffices for outcomes consistent with the underlying structure, though learning the network itself may be a different matter. Thus, in situations where regulators care only about behavior, they do not need to spend resources on helping members learn the entire network. However, in situations where agents must understand the network structure of which they are members, the relevant intervention is one that improves the separating power of equilibrium behavior, not one that expands the data agents see.

More broadly, the framework provides a tractable way to study environments in which the flow of information is itself uncertain. Many settings typically modeled with imperfect observability can be reinterpreted as problems of learning an underlying interaction structure. We have framed interactions as bilateral links, but the analysis uses only that payoffs and observations are governed by a common latent structure from a finite set. The same logic extends to richer structures, such as hypergraphs, multilayer relations, and networks with shadow links, with the speed of learning governed by the corresponding notion of distance. To sum up, the paper separates what agents need to behave optimally from what they need to know, shows that these two requirements need not coincide, and provides a framework for further study on partially known interaction structures.

Proofs

Proof of Lemma 1.

Fix a myopic Bayesian equilibrium σ^* and a player $i \in N$. Let $\Pi_i^\tau(\sigma^*) = \Pi_i^\tau$ be the partition of $\mathcal{G}_n$ induced by the equivalence relation $\sim_{i,\tau}$ under σ^*. Formally, $\Pi_i^\tau = \left\{[\mathbf{g}]_{i,\tau} : \mathbf{g} \in \mathcal{G}_n\right\}$. We first show that $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ.

For any $\tau \geq 1$ choose any equivalence class $[\mathbf{g}]_{i,\tau+1} \in \Pi_i^{\tau+1}$ arbitrarily and let $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$. Then, from the definition of histories we get,

$$h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*) \Rightarrow h_i^\tau(\mathbf{g}, \sigma^*) = h_i^\tau(\mathbf{g}', \sigma^*).$$

From the last equality it follows that, $\mathbf{g} \sim_{i,\tau} \mathbf{g}' \Rightarrow \mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Hence, for any $\mathbf{g} \in \mathcal{G}_n$ we have $[\mathbf{g}]_{i,\tau+1} \subseteq [\mathbf{g}]_{i,\tau}$. Therefore, $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ.

Let $b_i^\tau = |\Pi_i^\tau|$. Since $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ, the sequence $(b_i^\tau)_{\tau\geq 1}$ is weakly increasing. Also, because each Π_i^τ is a partition of the finite set $\mathcal{G}_n$, we have $1 \leq b_i^\tau \leq |\mathcal{G}_n|$ for every $\tau \geq 1$. Hence $(b_i^\tau)_{\tau\geq 1}$ is an integer-valued weakly increasing sequence that is bounded above. Therefore, there exists $T_i < \infty$ such that $b_i^\tau = b_i^{T_i}$ for all $\tau \geq T_i$.

Fix any $\tau \geq T_i$. Since $\Pi_i^{\tau+1}$ refines Π_i^τ and the two partitions have the same number of elements, this refinement cannot be strict. Thus $\Pi_i^{\tau+1} = \Pi_i^\tau$. Repeating the same argument period by period, we obtain $\Pi_i^\tau = \Pi_i^{T_i}$ for every $\tau \geq T_i$. Thus the equivalence class in the partition containing $\mathbf{g}$ is the same at periods τ and T_i. Therefore, $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,T_i}$ for every $\mathbf{g} \in \mathcal{G}_n$ and every $\tau \geq T_i$.

Since N is finite, let $T = \max_{i \in N} T_i$. Then $T < \infty$, and for every player i, every interaction structure $\mathbf{g} \in \mathcal{G}_n$, and every period $\tau \geq T$, we have $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,T}$. Thus no player's observational-equivalence class changes after period T. Therefore, learning ends in finite time. ■

Proof of Theorem 1.

Fix the myopic Bayesian equilibrium σ^*, a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$, a period $\tau \geq \tau^*$, and a player $i \in N$. Write h_i^τ for $h_i^\tau(\mathbf{g}, \sigma^*)$. Since $h_i^\tau = h_i^\tau(\mathbf{g}, \sigma^*)$ is generated by the realized interaction structure $\mathbf{g}$ under σ^*, and since the prior has full support, we have $P_{i,\sigma^*}(h_i^\tau) \geq p(\mathbf{g}) > 0$. Hence $h_i^\tau \in P_i(\sigma^*)$, so the posterior at h_i^τ is determined by Bayes' rule.

By Bayes' rule, player i assigns positive probability at h_i^τ exactly to those interaction structures that generate the same period-τ private history as $\mathbf{g}$. Thus $\mu_{i,\sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) > 0$ if and only if $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Moreover, because the first component of player i's private history is her observed neighborhood, every $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$ satisfies $N_i(\mathbf{g}') = N_i(\mathbf{g})$.

Now take any $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Since learning has ended at τ^* and $\tau \geq \tau^*$, the equivalence class of $\mathbf{g}$ for player i is the same at periods τ and $\tau + 1$. Hence $\mathbf{g}' \in [\mathbf{g}]_{i,\tau+1}$ as well. Therefore $h_i^{\tau+1}(\mathbf{g}', \sigma^*) = h_i^{\tau+1}(\mathbf{g}, \sigma^*)$. Since a period-$(\tau+1)$ history is obtained from the period-τ history by adding the period-τ signal, it follows that $z_i^\tau(\mathbf{g}', \sigma^*) = z_i^\tau(\mathbf{g}, \sigma^*)$. Together with $N_i(\mathbf{g}') = N_i(\mathbf{g})$, equality of signals implies that the actions observed by player i are the same under $\mathbf{g}'$ and under $\mathbf{g}$; that is, $a^\tau_{N_i(\mathbf{g}')}(\mathbf{g}', \sigma^*) = a^\tau_{N_i(\mathbf{g})}(\mathbf{g}, \sigma^*)$.

We can now compare player i's payoff across all interaction structures in the support of her posterior. Fix any $a_i \in A_i$ and any $\mathbf{g}'$ such that $\mu_{i,\sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) > 0$. From the previous paragraph, $\mathbf{g}'$ has the same neighborhood for player i as $\mathbf{g}$, and the corresponding local action vector is also the same. Therefore, by the local payoff-equivalence assumption,

$$u_i\Big(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}', \sigma^*)); \mathbf{g}'\Big) = u_i\Big(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g}\Big).$$

Thus every payoff term that receives positive posterior probability in player i's myopic objective is identical to the payoff she would obtain under the realized interaction structure $\mathbf{g}$.

Since the remaining interaction structures have zero posterior probability, player i's posterior expected payoff at h_i^τ is the same function of a_i as the complete-information payoff under the realized interaction structure $\mathbf{g}$, i.e.

$$\sum_{\mathbf{g}' \in \mathcal{G}_n} \mu_{i,\sigma^*}^\tau(\mathbf{g}' \mid h_i^\tau) u_i\Big(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}', \sigma^*)); \mathbf{g}'\Big) = u_i\Big(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g}\Big)$$

for every $a_i \in A_i$.

Therefore, by the myopic Bayesian equilibrium condition, $\sigma_i^{*\tau}(h_i^\tau)$ maximizes $u_i(a_i, \sigma_{-i}^{*\tau}(h_{-i}^\tau(\mathbf{g}, \sigma^*)); \mathbf{g})$ over A_i. Using the definition of induced play, this gives

$$a_i^\tau(\mathbf{g}, \sigma^*) \in \arg\max_{a_i \in A_i} u_i\big(a_i, a_{-i}^\tau(\mathbf{g}, \sigma^*); \mathbf{g}\big).$$

Hence $a_i^\tau(\mathbf{g}, \sigma^*)$ is a best response to $a_{-i}^\tau(\mathbf{g}, \sigma^*)$ in the complete-information stage game induced by $\mathbf{g}$.

Since this argument applies to every player $i \in N$, the realized action profile $a^\tau(\mathbf{g}, \sigma^*)$ is a Nash equilibrium of the complete-information stage game under $\mathbf{g}$. Therefore, for every $\tau \geq \tau^*$,

$$a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g}),$$

where $NE(\mathbf{g})$ denotes the set of pure Nash equilibria of the complete-information stage game induced by $\mathbf{g}$. ■

Proof of Theorem 2.

Fix the myopic Bayesian equilibrium σ^* and a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$. Suppose learning ends at τ^* under σ^*.

First suppose there exists $j \in N_i(\mathbf{g})$ such that $NE_j(\mathbf{g}') \cap NE_j(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$. If possible, let us assume that learning is not perfect for player i. Then there exists $\mathbf{g}' \neq \mathbf{g}$ such that $\mathbf{g}' \in [\mathbf{g}]_{i,\tau^*}$.

Since learning ends at τ^*, the equivalence class of $\mathbf{g}$ for player i is the same at periods τ^* and $\tau^* + 1$. Hence $\mathbf{g}' \in [\mathbf{g}]_{i,\tau^*+1}$ as well. Therefore $h_i^{\tau^*+1}(\mathbf{g}', \sigma^*) = h_i^{\tau^*+1}(\mathbf{g}, \sigma^*)$, and so the period-$\tau^*$ signals of player i coincide under $\mathbf{g}'$ and $\mathbf{g}$. Moreover, because $\mathbf{g}' \in [\mathbf{g}]_{i,\tau^*}$, the initial component of player i's history gives $N_i(\mathbf{g}') = N_i(\mathbf{g})$. Since $j \in N_i(\mathbf{g})$, equality of signals implies $a_j^{\tau^*}(\mathbf{g}', \sigma^*) = a_j^{\tau^*}(\mathbf{g}, \sigma^*)$.

By Theorem 1, applied to both interaction structures, $a^{\tau^*}(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ and $a^{\tau^*}(\mathbf{g}', \sigma^*) \in NE(\mathbf{g}')$. Hence,

$$a_j^{\tau^*}(\mathbf{g}, \sigma^*) = a_j^{\tau^*}(\mathbf{g}', \sigma^*) \in NE_j(\mathbf{g}) \cap NE_j(\mathbf{g}').$$

This contradicts the assumed separation of $NE_j(\cdot)$ across distinct interaction structures. Therefore $[\mathbf{g}]_{i,\tau^*} = \{\mathbf{g}\}$.

Now suppose instead that $NE_i(\mathbf{g}') \cap NE_i(\mathbf{g}'') = \emptyset$ for every distinct $\mathbf{g}', \mathbf{g}'' \in \mathcal{G}_n$. Again if possible, let us assume that learning is not perfect for player i, and choose $\mathbf{g}' \neq \mathbf{g}$ with $\mathbf{g}' \in [\mathbf{g}]_{i,\tau^*}$. Then $h_i^{\tau^*}(\mathbf{g}', \sigma^*) = h_i^{\tau^*}(\mathbf{g}, \sigma^*)$. Since σ_i^* is a pure strategy, equal private histories imply $a_i^{\tau^*}(\mathbf{g}', \sigma^*) = a_i^{\tau^*}(\mathbf{g}, \sigma^*)$.

By Theorem 1, $a^{\tau^*}(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ and $a^{\tau^*}(\mathbf{g}', \sigma^*) \in NE(\mathbf{g}')$. Hence,

$$a_i^{\tau^*}(\mathbf{g}', \sigma^*) = a_i^{\tau^*}(\mathbf{g}, \sigma^*) \in NE_i(\mathbf{g}) \cap NE_i(\mathbf{g}'),$$

contradicting the assumed separation of $NE_i(\cdot)$ across distinct interaction structures. Thus $[\mathbf{g}]_{i,\tau^*} = \{\mathbf{g}\}$. ■

Proof of Corollary 1.

Fix a myopic Bayesian equilibrium σ^* and a realized interaction structure $\mathbf{g} \in \mathcal{G}_n$. Suppose learning ends at period τ^* under σ^*.

By Theorem 1, for every $\tau \geq \tau^*$, $a^{\tau}(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$. It remains to characterize the complete-information Nash equilibrium set for the linear-quadratic stage game.

Under (1), player i's payoff is strictly concave in a_i, and hence the first-order condition is necessary and sufficient. Thus any complete-information Nash equilibrium a under $\mathbf{g}$ satisfies $1 - a_i + \lambda \sum_{j \in N} g_{ij}(\mathbf{g}) a_j = 0$ for every $i \in N$. Stacking these equations gives

$$\big(I - \lambda G(\mathbf{g})\big)a = \mathbf{1}.$$

Since $0 < \lambda < 1/\bar{\rho}$ and $\rho(G(\mathbf{g})) \leq \bar{\rho}$, the matrix $I - \lambda G(\mathbf{g})$ is invertible. Therefore the complete-information Nash equilibrium under $\mathbf{g}$ is unique and is given by

$$a^c(\mathbf{g}) = \big(I - \lambda G(\mathbf{g})\big)^{-1}\mathbf{1}.$$

Since $a^\tau(\mathbf{g}, \sigma^*) \in NE(\mathbf{g})$ for every $\tau \geq \tau^*$, uniqueness implies

$$a_i^\tau(\mathbf{g}, \sigma^*) = \Big[\big(I - \lambda G(\mathbf{g})\big)^{-1}\mathbf{1}\Big]_i \qquad \text{for every } i \in N \text{ and every } \tau \geq \tau^*.$$

■

Proof of Proposition 1.

Fix $i \in N$ and two interaction structures $\mathbf{g}, \mathbf{g}' \in \mathcal{G}_n$. Write $G := G(\mathbf{g})$ and $G' := G(\mathbf{g}')$, and let $\rho^* := \max\{\rho(G), \rho(G')\}$. For $\lambda \in (0, 1/\rho^*)$, define

$$f_i(\lambda) := e_i^\top (I - \lambda G)^{-1}\mathbf{1} - e_i^\top (I - \lambda G')^{-1}\mathbf{1}.$$

Thus $f_i(\lambda) = a_i^c(\mathbf{g}) - a_i^c(\mathbf{g}')$ whenever the two complete-information equilibrium vectors are well defined.

Since $\lambda\rho(G) < 1$ and $\lambda\rho(G') < 1$ for every $\lambda \in (0, 1/\rho^*)$, both Neumann-series expansions are valid. Hence

$$f_i(\lambda) = \sum_{m=0}^{\infty} \lambda^m \left[(G^m\mathbf{1})_i - ((G')^m\mathbf{1})_i\right] = \sum_{m=0}^{\infty} \lambda^m \left[W_i^m(\mathbf{g}) - W_i^m(\mathbf{g}')\right].$$

By assumption, there exists $k \geq 1$ such that $W_i^k(\mathbf{g}) \neq W_i^k(\mathbf{g}')$. Therefore the above power series has a nonzero coefficient, so f_i is not identically zero on $(0, 1/\rho^*)$.

It remains to rule out infinitely many coincidences. Each coordinate of $(I - \lambda G)^{-1}\mathbf{1}$ is a rational function of λ, since

$$e_i^\top (I - \lambda G)^{-1}\mathbf{1} = \frac{P_i(\lambda)}{\det(I - \lambda G)}$$

for some polynomial P_i, and similarly for G'. Thus f_i can be written as a rational function $N_i(\lambda)/D_i(\lambda)$, where $D_i(\lambda) \neq 0$ for all $\lambda \in (0, 1/\rho^*)$. Since f_i is not identically zero, N_i is not the zero polynomial. Hence N_i has only finitely many roots.

Therefore there exists a finite and hence null set, $\Lambda_i(\mathbf{g}, \mathbf{g}') \subset (0, 1/\rho^*)$ such that $f_i(\lambda) \neq 0$ for every $\lambda \in (0, 1/\rho^*) \setminus \Lambda_i(\mathbf{g}, \mathbf{g}')$. Equivalently,

$$a_i^c(\mathbf{g}) \neq a_i^c(\mathbf{g}') \qquad \text{for all } \lambda \in (0, 1/\rho^*) \setminus \Lambda_i(\mathbf{g}, \mathbf{g}').$$

■

Proof of Corollary 2.

Fix $\mathbf{g} \in \mathcal{G}_n$ and $i \in N$. Suppose there exists $j \in \{i\} \cup N_i(\mathbf{g})$ such that the walk profile $\mathbf{h} \mapsto (W_j^k(\mathbf{h}))_{k \geq 0}$ distinguishes every pair of interaction structures in $\mathcal{G}_n$.

Take any two distinct interaction structures $\mathbf{h}, \mathbf{h}' \in \mathcal{G}_n$. By assumption, there exists $k \geq 0$ such that $W_j^k(\mathbf{h}) \neq W_j^k(\mathbf{h}')$. Since $W_j^0(\mathbf{h}) = W_j^0(\mathbf{h}') = 1$, this difference must occur at some $k \geq 1$. Proposition 1 then implies that there is a finite and hence null set, $\Lambda_j(\mathbf{h}, \mathbf{h}') \subset (0, 1/\bar{\rho})$ such that $a_j^c(\mathbf{h}) \neq a_j^c(\mathbf{h}')$ for every $\lambda \in (0, 1/\bar{\rho}) \setminus \Lambda_j(\mathbf{h}, \mathbf{h}')$.

Since $\mathcal{G}_n$ is finite, the union $\Lambda_j := \bigcup_{\mathbf{h} \neq \mathbf{h}'} \Lambda_j(\mathbf{h}, \mathbf{h}')$ is finite. Hence, for every $\lambda \in (0, 1/\bar{\rho}) \setminus \Lambda_j$, the map $\mathbf{h} \mapsto a_j^c(\mathbf{h})$ is injective on $\mathcal{G}_n$.

In the linear-quadratic stage game, the complete-information Nash equilibrium under each $\mathbf{h}$ is unique. Therefore $NE_j(\mathbf{h}) = \{a_j^c(\mathbf{h})\}$ for every $\mathbf{h} \in \mathcal{G}_n$. Thus, for every admissible $\lambda \notin \Lambda_j$, the equilibrium sets $NE_j(\mathbf{h})$ are pairwise disjoint across distinct interaction structures.

If $j = i$, condition (i) of Theorem 2 applies. If $j \in N_i(\mathbf{g})$, condition (ii) of Theorem 2 applies. Therefore, whenever learning terminates at τ^* under σ^*, player i learns the realized interaction structure perfectly: $[\mathbf{g}]_{i,\tau^*} = \{\mathbf{g}\}$. ■

Proof of Theorem 3.

Write $\Pi_i^\tau = \Pi_i^\tau(\sigma^*)$ and $b_i^\tau = b_i^\tau(\sigma^*)$. Fix a player $i \in N$. Since Π_i^τ is a partition of the finite set $\mathcal{G}_n$, we have $1 \leq b_i^\tau \leq |\mathcal{G}_n|$ for every $\tau \geq 1$. Moreover, since $\Pi_i^{\tau+1}$ refines Π_i^τ, the sequence $(b_i^\tau)_{\tau \geq 1}$ is weakly increasing. Hence, for each player i, this sequence can strictly increase at most $|\mathcal{G}_n| - 1$ times.

We now show that, before learning ends, some player's partition must strictly refine in every period. Fix any period τ. Suppose, toward a contradiction, that no player's partition strictly refines from τ to $\tau+1$. Since each $\Pi_i^{\tau+1}$ refines Π_i^τ, this means $\Pi_i^{\tau+1} = \Pi_i^\tau$, for every $i \in N$. By the partition-stationarity condition,

$$a_i^{\tau+1}(\mathbf{g}, \sigma^*) = a_i^\tau(\mathbf{g}, \sigma^*) \qquad \text{for every } i \in N \text{ and every } \mathbf{g} \in \mathcal{G}_n.$$

Therefore, the period-$(\tau+1)$ action profile carries exactly the same graph-dependent information as the period-τ action profile. Since the period-τ action profile did not refine any player's partition, the period-$(\tau+1)$ action profile cannot refine any player's partition either. Hence $\Pi_i^{\tau+2} = \Pi_i^{\tau+1}$, for every $i \in N$. Repeating the same argument inductively gives

$$\Pi_i^s = \Pi_i^\tau \qquad \text{for every } s \geq \tau \text{ and every } i \in N.$$

Thus learning has already stabilized at period τ.

Therefore, if $\tau < \tau^*(\sigma^*)$, it cannot be the case that all players' partitions remain unchanged from τ to $\tau+1$. Hence there must exist some player $i \in N$ and some interaction structure $\mathbf{g} \in \mathcal{G}_n$ such that $[\mathbf{g}]_{i,\tau+1} \subset [\mathbf{g}]_{i,\tau}$. Since $\Pi_i^{\tau+1}$ is a refinement of Π_i^τ, this strict containment means that the refinement is strict, and therefore $b_i^{\tau+1} > b_i^\tau$.

Thus every period before $\tau^*(\sigma^*)$ generates at least one strict increase in one of the n sequences $(b_i^\tau)_{\tau\geq 1}$, $i \in N$. Since each such sequence can strictly increase at most $|\mathcal{G}_n|-1$ times, the total number of strict increases across all players is at most $n(|\mathcal{G}_n|-1)$. Hence the number of periods before $\tau^*(\sigma^*)$ is at most $n(|\mathcal{G}_n|-1)$, and so

$$\tau^*(\sigma^*) \leq 1 + n(|\mathcal{G}_n| - 1).$$

$\blacksquare$

Proof of Lemma 2.

We prove the two implications by induction on τ.

For $\tau = 1$, the claim follows directly from the definitions. Indeed, $h_i^1(\mathbf{g}, \sigma^*) = N_i(\mathbf{g})$, while $R_i^1(\mathbf{g})$ records exactly $N_i(\mathbf{g})$ since $N_i^0(\mathbf{g}) = \{i\}$. Hence $R_i^1(\mathbf{g}) = R_i^1(\mathbf{g}')$ if and only if $h_i^1(\mathbf{g}, \sigma^*) = h_i^1(\mathbf{g}', \sigma^*)$. Thus both implications hold at $\tau = 1$.

Fix $\tau \geq 1$ and suppose the result holds at period τ for every player. We first prove the forward implication at period $\tau + 1$. Suppose $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$. Then, in particular, $N_i(\mathbf{g}) = N_i(\mathbf{g}')$. For every $j \in N_i(\mathbf{g}) \cup \{i\}$ and every $\ell \in N_j^{\tau-1}(\mathbf{g})$,

$$\mathrm{dist}_{\mathbf{g}}(\ell, i) \leq \mathrm{dist}_{\mathbf{g}}(\ell, j) + \mathrm{dist}_{\mathbf{g}}(j, i) \leq \tau.$$

Hence $N_j^{\tau-1}(\mathbf{g}) \subseteq N_i^{\tau}(\mathbf{g})$. Since

$$R_i^{\tau+1}(\mathbf{g}) = \big(N_\ell(\mathbf{g})\big)_{\ell \in N_i^{\tau}(\mathbf{g})} \quad \text{and} \quad R_j^{\tau}(\mathbf{g}) = \big(N_\ell(\mathbf{g})\big)_{\ell \in N_j^{\tau-1}(\mathbf{g})},$$

the latter is determined by the former. Therefore $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$ implies $R_j^{\tau}(\mathbf{g}) = R_j^{\tau}(\mathbf{g}')$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. By the induction hypothesis, $h_j^{\tau}(\mathbf{g}, \sigma^*) = h_j^{\tau}(\mathbf{g}', \sigma^*)$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. Since strategies are pure, these equal histories imply equal period-τ actions for all those players. Together with $N_i(\mathbf{g}) = N_i(\mathbf{g}')$, this gives $z_i^{\tau}(\mathbf{g}, \sigma^*) = z_i^{\tau}(\mathbf{g}', \sigma^*)$. Since the period-$\tau$ histories of player i are also equal by the induction hypothesis, the recursive definition of histories gives $h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*)$.

We now prove the converse at period $\tau + 1$ under the separation condition. Suppose $h_i^{\tau+1}(\mathbf{g}, \sigma^*) = h_i^{\tau+1}(\mathbf{g}', \sigma^*)$. Then both the period-$\tau$ histories and the period-τ signals of player i coincide under $\mathbf{g}$ and $\mathbf{g}'$. In particular, $N_i(\mathbf{g}) = N_i(\mathbf{g}')$, and equality of signals gives $a_j^{\tau}(\mathbf{g}, \sigma^*) = a_j^{\tau}(\mathbf{g}', \sigma^*)$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$.

By the contrapositive of separation condition, these equal actions imply $[\mathbf{g}]_{j,\tau} = [\mathbf{g}']_{j,\tau}$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$. Hence $h_j^{\tau}(\mathbf{g}, \sigma^*) = h_j^{\tau}(\mathbf{g}', \sigma^*)$ for every such j. Applying the induction hypothesis for the converse implication, we obtain $R_j^{\tau}(\mathbf{g}) = R_j^{\tau}(\mathbf{g}')$ for every $j \in N_i(\mathbf{g}) \cup \{i\}$.

Finally, $R_i^{\tau+1}(\mathbf{g})$ is determined by $N_i(\mathbf{g})$ together with the collection $\{R_j^{\tau}(\mathbf{g}) : j \in N_i(\mathbf{g}) \cup \{i\}\}$. Since each of these objects is the same under $\mathbf{g}$ and $\mathbf{g}'$, it follows that $R_i^{\tau+1}(\mathbf{g}) = R_i^{\tau+1}(\mathbf{g}')$.

By induction, the result holds for every $\tau \geq 1$. ■

Proof of Proposition 2.

Suppose the separation condition holds and let $D := d(\mathcal{G}_n)$. Fix $i \in N$, $\mathbf{g} \in \mathcal{G}_n$, and $\tau \geq D+1$. We show that $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,D+1}$.

Take any $\mathbf{g}' \in [\mathbf{g}]_{i,D+1}$. Then $h_i^{D+1}(\mathbf{g}', \sigma^*) = h_i^{D+1}(\mathbf{g}, \sigma^*)$. By Lemma 2 and the separation condition, this implies $R_i^{D+1}(\mathbf{g}') = R_i^{D+1}(\mathbf{g})$.

Now fix an arbitrary $\tilde{\mathbf{g}} \in \mathcal{G}_n$. By definition of D, every node that reaches i under $\tilde{\mathbf{g}}$ does so along a path of length at most D. Hence $N_i^r(\tilde{\mathbf{g}}) = N_i^D(\tilde{\mathbf{g}})$ for every $r \geq D$. Therefore, for every $\tau \geq D+1$,

$$R_i^\tau(\tilde{\mathbf{g}}) = \big(N_j(\tilde{\mathbf{g}})\big)_{j \in N_i^{\tau-1}(\tilde{\mathbf{g}})} = \big(N_j(\tilde{\mathbf{g}})\big)_{j \in N_i^{D}(\tilde{\mathbf{g}})} = R_i^{D+1}(\tilde{\mathbf{g}}).$$

In particular,

$$R_i^\tau(\mathbf{g}') = R_i^{D+1}(\mathbf{g}') = R_i^{D+1}(\mathbf{g}) = R_i^\tau(\mathbf{g}).$$

Lemma 2 then gives $h_i^\tau(\mathbf{g}', \sigma^*) = h_i^\tau(\mathbf{g}, \sigma^*)$, so $\mathbf{g}' \in [\mathbf{g}]_{i,\tau}$. Hence $[\mathbf{g}]_{i,D+1} \subseteq [\mathbf{g}]_{i,\tau}$.

The reverse inclusion follows from the fact that histories are nested over time: if two structures generate the same period-τ history for player i, then they also generate the same period-$(D+1)$ history since $\tau \geq D+1$. Thus $[\mathbf{g}]_{i,\tau} \subseteq [\mathbf{g}]_{i,D+1}$. Therefore $[\mathbf{g}]_{i,\tau} = [\mathbf{g}]_{i,D+1}$ for every $\tau \geq D+1$.

Since the argument is uniform over i and $\mathbf{g}$, learning stabilizes by period $D+1$. Hence

$$\tau^*(\sigma^*) \leq D+1 = d(\mathcal{G}_n) + 1.$$

■

Proof of Corollary 3.

Fix any $\mathbf{g} \in \mathcal{G}_n$ and any players i, j with

$$\mathrm{dist}_{\mathbf{g}}(j, i) < \infty.$$

By definition, $\mathrm{dist}_{\mathbf{g}}(j, i)$ is the length of a shortest directed path from j to i. Any shortest directed path is simple, meaning that it visits each player at most once., since otherwise one could delete a directed cycle and obtain a shorter directed path from j to i. A simple directed path on n players has length at most $n-1$. Thus

$$\mathrm{dist}_{\mathbf{g}}(j,i) \leq n-1.$$

Hence

$$d(\mathcal{G}_n) = \max_{\mathbf{g}\in\mathcal{G}_n} \mathrm{diam}(\mathbf{g}) \leq n-1.$$

Applying Proposition 2, we obtain

$$\tau^*(\sigma^*) \leq d(\mathcal{G}_n) + 1 \leq n.$$

■

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